Running Docker Apps in the AWS Cloud

Dominic Zarro (ADNET Systems, Inc.)

SESDA Data Center CIG
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Outline

- Setting up an AWS EC2 instance
- Running Docker
- Running a Docker Python app
- Distributing a Docker image

Setting up AWS EC2 instance

1. Apply for 12 month free-tier account:

   https://aws.amazon.com/free

2. Create Elastic Compute Cloud (EC2) instance:

   https://aws.amazon.com/ec2

   - Up to 750 hours per month of Linux (or Windows) server uptime with 1 GB memory
   - Up to 30 GB of permanent block storage
   - Up to15 GB of bandwidth

Docker

- What is Docker?
  - Docker is a platform to develop, deploy, and run applications within containers:

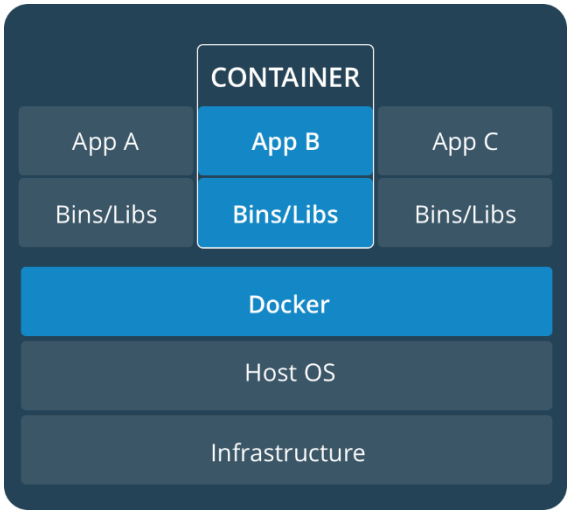
    https://docs.docker.com/get-started

- What is a Docker Container?
  - A container is a running Docker image.

- What is a Docker Image?
  - An image is an executable package that includes everything needed to run an application--the code, a runtime, libraries, environment variables, and configuration files.

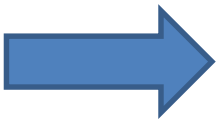
Docker Background

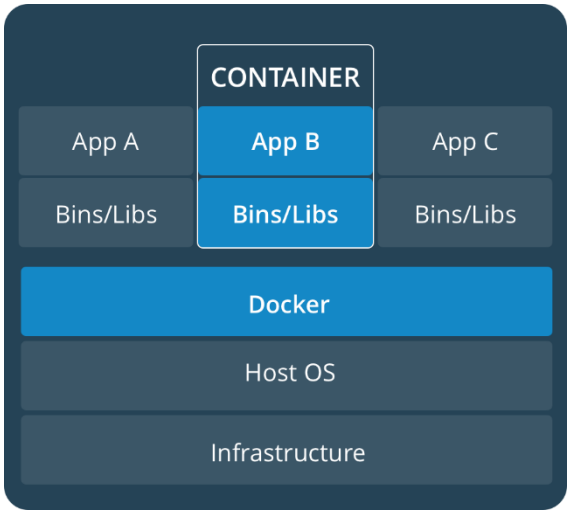
- Why use Docker?
  - Flexible: Complex applications can be containerized.
  - Lightweight: Containers leverage and share the host kernel.
  - Portable: Build images locally, deploy to the cloud, and run anywhere.
  - Distributed: Access Docker Hub registry for pre-built and tested images: https://hub.docker.com



Local Host

Export App

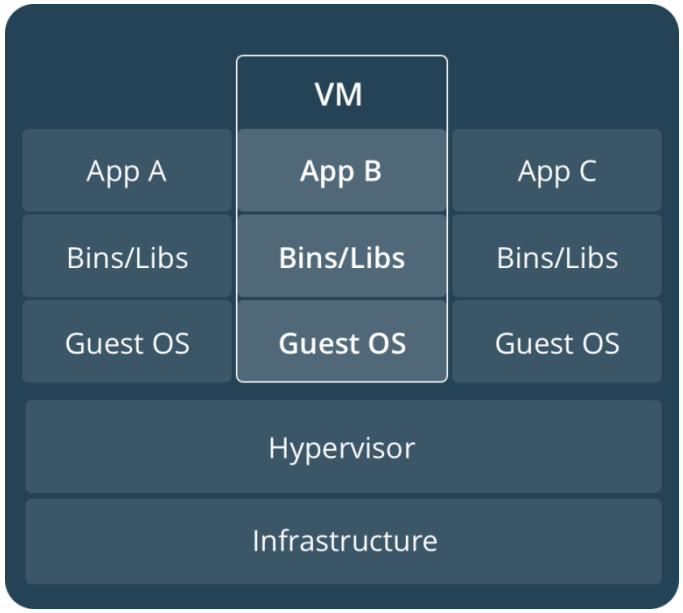




Remote Host

Docker Background

- How does Docker differ from a virtual machine (VM)?
  - VM runs full-blown “guest” operating system with virtual access to host resources.
  - VM provides environment with more resources than most applications need.



Docker Example

- Amazon Linux has Docker pre-installed

```
$ docker info
 Containers: 42
 Running: 0
 Paused: 0
 Stopped: 42
 Images: 25
 Server Version: 18.03.1-ce
```

- Run Hello World (pre-built image from Docker Hub)

```
$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
9bb5a5d4561a: Already exists
Digest:
sha256:f5233545e43561214ca4891fd1157e1c3c563316ed8e237750d59bde73361e77
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
```

Build a Docker Image

- Create a Dockerfile (simple instruction text)

```
FROM python:3      # pull latest Python 3 image from Hub
CMD ["python"]     # run this command when running image
```

- Build Docker Image (in Dockerfile directory)

```
$ docker build –t python3 .
   Sending build context to Docker daemon  8.192kB
   Step 1/2 : FROM python:3
    ---> a5b7afcfdcc8
   Step 2/2 : CMD [ "python"]
    ---> Running in e612cb3c4d5a
   Removing intermediate container e612cb3c4d5a
    ---> 697d84845520
   Successfully built 697d84845520
   Successfully tagged python3:latest
```

Run Docker Image in Container

- Verify image was created

```
$ docker images

REPOSITORY          TAG                 IMAGE ID            CREATED          SIZE
python3             latest              d4d98018e083        3 minutes ago       912MB
```

- Run image in an interactive terminal container

```
$ docker run –it python3

Python 3.6.5 (default, Jun  6 2018, 19:19:24)
[GCC 6.3.0 20170516] on linux
Type "help", "copyright", "credits" or "license" for more
information.
>>>
>> print("Hello World")
Hello World
>>>
```

Add Display Functionality to Image

- Add Run command to Dockerfile to install Matplotlib in image

```
FROM python:3                   # pull latest Python3 image from cache
RUN pip install matplotlib      # install matplotlib
CMD ["python"]                  # run this command when running image
```

- Re-build Python 3 image

```
$ docker build -t python3 .
```

- Re-run image in container (need X-forwarding for DISPLAY)

```
$ docker run -it  \
  -v /tmp/.X11-unix \  # need these lines to forward X-display
  -e DISPLAY \
  python3
```

Plot Python Numpy Array

```
Python 3.6.5 (default, Jun  6 2018, 19:19:24)
[GCC 6.3.0 20170516] on linux
Type "help", "copyright", "credits" or "license" for more
information.
>>>
>>> import matplotlib.pyplot as plt
>>> import numpy as np
>>> x=np.arange(0.0, 5.0, 0.1)
>>> y=np.cos(x*np.pi)
>>> plt.plot(x,y)
>>> plt.show()
```

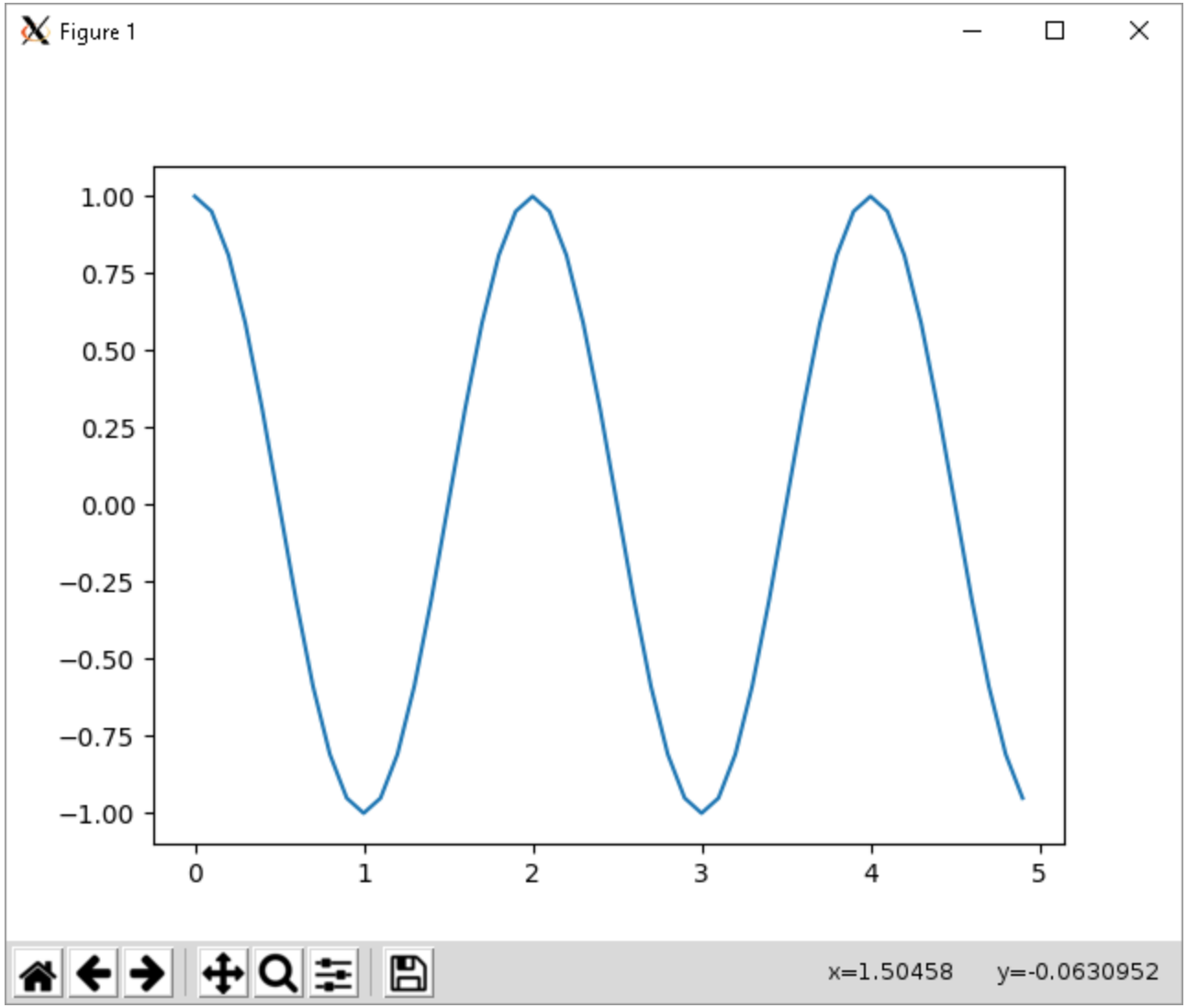


Figure 1
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
1
2
3
4
5
x=1.50458
y=-0.0630952

Run External Python Function

- Mount local directory as external disk volume

```
$ docker run -it  \
  -v ${PWD}:/home/python  # mount current directory inside container
  -v /tmp/.X11-unix \
  -e DISPLAY \
  python3
```

- Import Python library module from local directory

```
Python 3.6.5 (default, Jun  6 2018, 19:19:24)
[GCC 6.3.0 20170516] on linux
Type "help", "copyright", "credits" or "license" for more
information.
>>>
>>> import myplot
>>> myplot.run()
```

Distribute a Docker Image

- Set up account on DockerHub:

  https://hub.docker.com

- Log into DockerHub:

```
$ docker login
```

- Tag image with username:

```
$ docker tag python3 dzarro/python3
```

- Push image to DockerHub:

```
$ docker push dzarro/python3
```

- Pull image to new host:

```
$ docker pull dzarro/python3
```